

ML for finance

Final exam

February 19, 2026

The duration of the exam is **two hours** (2:00). Using ChatGPT is not allowed. Communicating with other students is not allowed. The exercises are independent. With this *pdf* file, you will find the following files : *pisa.csv*, *GDP-US.csv*, *HouseholdDebt-US.csv*, *NonFinancialDebt-US.csv*, *PublicDebt-US.csv*, *3MtreasuryBills-US.csv*, *Yield10years-US.csv*, *IndexTotalSharePrices-US.csv*, *forGrowth.py*. Please comment your code. At the end of the exam, you have to send your files (zipped) by mail.

1 Predicting PISA score

The *pisa.csv* file contains the PISA scores of 50 countries as well as the values of six numerical variables that can influence these scores. You are asked to write a Python script to perform the tasks listed below.

1. Read the data.
2. Represent the *Score* column as a histogram.
3. Add a *Score2* column that contains only two values: 1 for scores above a certain value *m* and 0 for the others.
4. Create a decision tree for *Score2* based on the other variables.
5. Calculate the accuracy of this tree and give its confusion matrix.
6. Represent this tree graphically and list its rules.

2 Predicting the GDP growth

In this exercise, our objective is to create and analyze a classifier (first a random forest then a neural network) for forecasting (the class of) GDP growth.

2.1 Discovering the data

The data is created by combining in one dataframe (*myData2*) the history of several indicators : the GDP value, the household debt to GDP rate, the non financial sector debt to GDP rate, the public debt to GDP rate, the 3-month treasury bill, the market yield on U.S. treasury securities at 10-year, and the index of total share prices. The code corresponding to this part of the work is in the file *forGrowth.py*.

1. Read the file *forGrowth.py* and add a code showing the number of rows and the number and the names of the columns of the dataframe *myData2*.
2. Add a column *Cls2* containing 2 possible values : Low and High (depending on the values of GDP growth).
3. Add a column *Cls3* containing 3 possible values : Low, Intermediate and High (depending on the values of GDP growth).

2.2 Random forest

1. Create random forests with different values of the number of trees, for predicting *Cls2* in function of the explanatory variables. Show the accuracy and the OOB error for each random forest.
2. For the random forest giving the best accuracy, plot the importance of the variables.
3. Create a new random forest with a selected set of variables (the most important) and give its accuracy and the OOB error.
4. Show the confusion matrix of this RF. Which class is the most difficult to predict ?
5. Extract and show one decision tree from the random forest.
6. Create and evaluate a new random forest using the most important variables to predict *Cls3*.

7. Give a general conclusion

2.3 neural network

1. Create and train a neural network for predicting Cls3 in function of the explanatory variables.
2. Evaluate the neural network.
3. Give the minimum and maximum values of each explanatory variable.
4. How can we improve the performance of our neural network ?
5. Create and train a new neural network and evaluate it.
6. Create, train and evaluate a new neural network for predicting Cls2.
7. Give a general conclusion.